

Improving the Quality of Pharmacotherapy in Elderly Primary Care Patients Through Medication Reviews: A Randomised Controlled Study

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Published online: 14 February 2013
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Abstract

Background Polypharmacy in the Swedish elderly population is currently a prioritised area of research with a focus on reducing the use of potentially inappropriate medications (PIMs). Multi-professional interventions have previously been tested for their ability to improve drug therapy in frail elderly patients.

Objective This study aimed to assess a structured model for pharmacist-led medication reviews in primary health care in southern Sweden and to measure its effects on numbers of patients with PIMs (using the definition of the Swedish National Board of Health and Welfare) using ≥ 10 drugs and using ≥ 3 psychotropics.

Methods This study was a randomised controlled clinical trial performed in a group of patients aged ≥ 75 years and living in nursing homes or the community and receiving

municipal health care. Medication reviews were performed by trained clinical pharmacists based on nurse-initiated symptom assessments with team-based or distance feedback to the physician. Data were collected from the patients' electronic medication lists and medical records at baseline and 2 months after the medication review.

Results A total of 369 patients were included: 182 in the intervention group and 187 in the control group. One-third of the patients in both groups had at least one PIM at baseline. Two months after the medication reviews, the number of intervention group patients with at least one PIM and the number of intervention group patients using ten or more drugs had decreased ($p = 0.007$ and $p = 0.001$, respectively), while there were no statistically significant changes in the control patients. No changes were seen in the number of patients using three or more

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psychotropic drugs, although the dosages of these drugs tended to decrease. Drug-related problems (DRPs) were identified in 93 % of the 182 patients in the intervention group. In total, there were 431 DRPs in the intervention group (a mean of 2.5 DRPs per patient, range 0–9, SD 1.5 at 95 % CI) and 16 % of the DRPs were related to PIMs.

Conclusions Medication reviews involving pharmacists in primary health care appear to be a feasible method to reduce the number of patients with PIMs, thus improving the quality of pharmacotherapy in elderly patients.

1 Background

1.1 The Challenge of Drug Therapy in the Elderly

The elderly population is increasing worldwide, and statistical demographic data estimate that 20 % of the global population will be older than 65 by 2025 [1]. According to the Swedish Central Bureau of Statistics, the proportion of the population aged 65 years or older was 18.8 % in Sweden in 2011. Aging is known to be associated with an increased prevalence of multiple chronic diseases and therefore the use of complex therapeutic regimes. Age-related changes in pharmacokinetics and pharmacodynamics [2], together with co-morbidity and polypharmacy, make the elderly a special group of patients who need to be treated with increased attention [1].

Polypharmacy is a controversial issue and has been found to be related to an increased risk of drug–drug interactions, higher morbidity in the older population, higher numbers of hospital admissions, lower compliance and increased institutionalisation [3]. A comprehensive literature review on the topic shows that polypharmacy is increasing in the elderly and is a major cause of morbidity and mortality in the elderly population worldwide [4]. Lack of continuity in physician contacts, lack of a consistent drug list, and inadequate prescribing and monitoring of drug therapy are some of the reasons for drug-related problems and the need for emergency hospital contacts [4]. A drug-related problem (DRP) has previously been described as “an undesirable patient experience that involves drug therapy and that actually or potentially interferes with a desired patient outcome” [5].

1.2 Potentially Inappropriate Medication

Well-defined criteria (Beers’ criteria) for potentially inappropriate medications (PIMs) in the elderly that use toxicological aspects and risk of adverse drug reactions have been described and were updated in 2012 [6]. The lack of good nationally adapted alternatives has led to the wide use in studies of the internationally accepted definition criteria

in order to create tools for identifying PIMs. About half of the drugs listed as PIMs in the Beers criteria are, however, unavailable in Europe. Therefore, criteria corresponding to European drug formularies have been developed, such as the Swedish quality indicators developed by the Swedish National Board of Health and Welfare [7]. They can work as a support for the prescriber in choosing appropriate medications but can even be used by drug and therapeutics committees to follow up doctors’ prescribing habits or to assess the quality of prescribing at the local or national level.

A nationwide register-based study in Sweden showed a strong correlation between the number of prescribed drugs and the number of PIMs, such as anticholinergic drugs, long-acting benzodiazepines, and three or more psychotropic drugs [8]. Use of multiple psychoactive drugs has been identified as particularly problematic in nursing home patients [9].

1.3 Medication Review

Optimisation of drug therapy in the elderly can be challenging, and different tools have been tested, such as educational outreach visits [10], medication reports at hospital discharge [11] and pharmaceutical care programmes using community pharmacists and medication reviews [12].

Currently, there is no well-established definition of the term “medication review” but Pharmaceutical Care Network Europe has suggested the following definition: “Medication review is an evaluation of patients’ medicines with the aim of managing the risk and optimising the outcome of medicine therapy by detecting, solving and preventing drug-related problems” [13].

Collaboration between physicians and pharmacists to identify drug-related problems has proven to be useful and led to better patient safety, as well as cost savings [14, 15]. Multi-disciplinary approaches have proved to be very satisfactory in the elderly patient, being appreciated by physicians and nurses, and had long-term effects on the patient’s drug therapy [16].

1.4 Multi-dose Drug Dispensing

Community-dwelling elderly individuals and nursing home residents in Sweden use on average eight to ten different drugs [7]. A large proportion of them use multi-dose drug dispensing (MDD). The goal of MDD is to create safer drug therapy, improve the patients’ drug management and adherence, get a complete picture of the patient’s drug prescriptions from different health-care providers as well as to improve communication between hospitals, primary care and communities. However, this service is used

primarily in Sweden and there are no studies to support evidence for such positive effects compared to traditional prescribing. According to data from 2005, 19 % of women and 13 % of men aged ≥ 75 years use MDD [8] and a majority of them live in nursing homes. The same study showed that 40 % of these patients were treated with at least one PIM. However, MDD led to fewer dangerous drug–drug interactions and may thus have advantages if used optimally. Disadvantages, including managing difficulties and uncritical renewal of prescriptions, have been mentioned [17]. A majority of the nursing home patients and community-dwelling patients with municipally provided home care in Sweden receive MDD because of high age, co-morbidity, cognitive impairment, polypharmacy and therefore increased care need. The medication is dispensed to the patient by the nurse and the intake is documented, leading to a high level of compliance.

1.5 The Medication Review in Primary Care in Southern Sweden

An integrated approach in which pharmacists help in the clinical routine has been developed in hospital care in Skåne County in southern Sweden (the Lund Integrated Medicines Management [LIMM] model) [18] and has been shown to reduce PIMs and drug-related hospital admissions [11]. This model of medication reviews for elderly patients with multiple illnesses originates from an early Swedish study in nursing homes, where medication reviews including the pharmacist in the multidisciplinary team produced a significant reduction in the number of psychotropic drugs [19]. In primary health care in Skåne County, medication reviews have been conducted during the past 10 years in different projects, both in nursing homes and community-dwelling elderly patients with multiple illnesses, and several models and approaches have been tried. The goal of medication reviews has been improved patient safety and quality of medication use, according to the Swedish National Board of Health and Welfare's indicators for good drug therapy in the elderly [7]. The instruments used in the LIMM model have been adapted to work in primary care. The main aim of adapting the instruments for primary care was to implement a new model of care with medication reviews before the patient's annual visit in order to improve the quality of elderly patients' pharmacotherapy in both community-dwelling and nursing home patients.

2 Objectives

The primary objective was to assess a structured model of care by studying the impact of pharmacist-led medication

reviews on the number of the patients using PIMs. Secondary objectives were to assess if this intervention model led to a decreased number of patients using ≥ 10 drugs and ≥ 3 psychotropics. The study also intended to classify and describe the types of DRPs identified during the intervention period and the medication reviews' impact on the patients' medication therapy.

3 Methods

The study received ethical approval from the Regional Ethical Review Board in Lund (no: 2011/245).

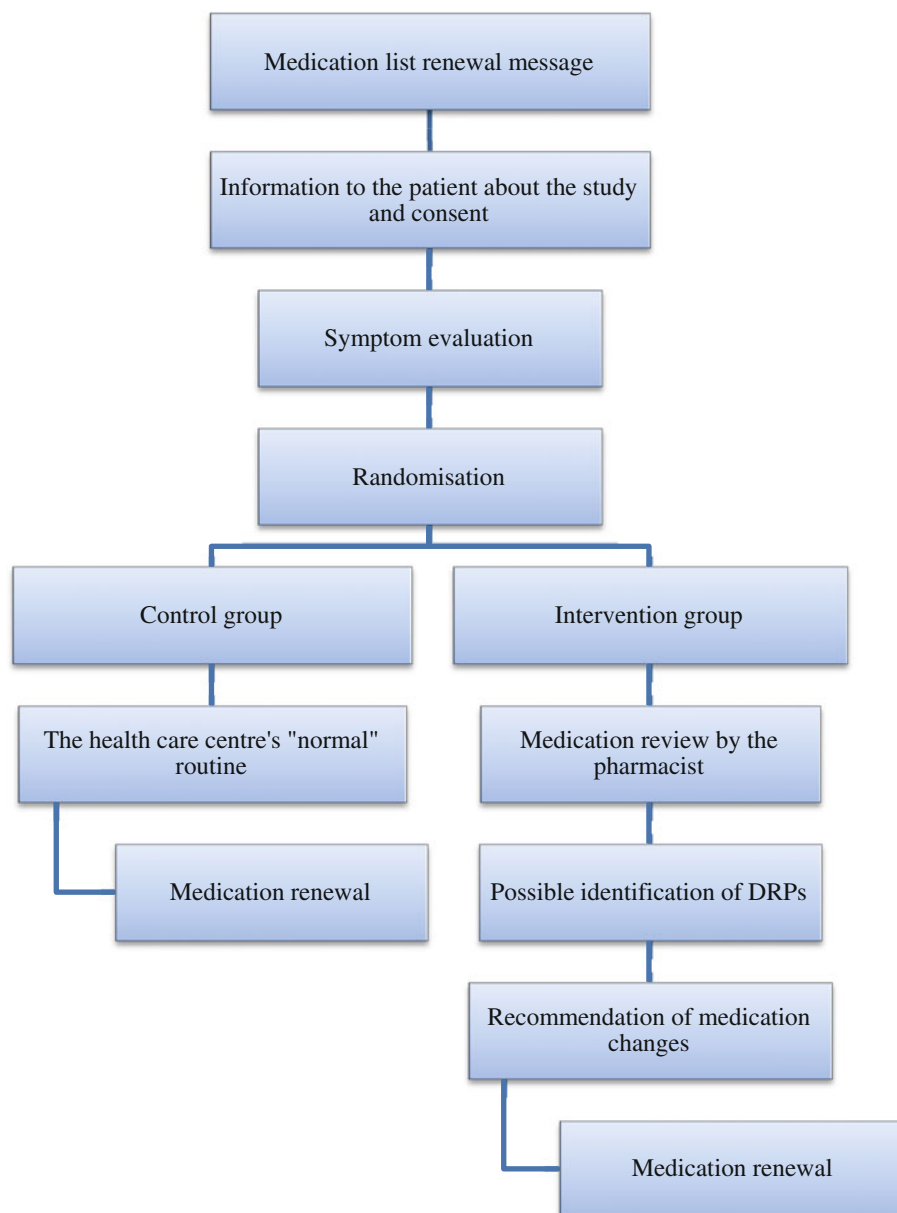
3.1 Study Setting and Design

Skåne County is situated in the southern part of Sweden and has approximately 1,150,000 inhabitants. Primary care is provided by public or private primary health care centres (PHCCs). There are 90 public and approximately 40 private PHCCs in Skåne. Due to practical reasons, such as to minimise the number of different electronic medical records (EMRs) we invited all public PHCCs to participate in this study. Four pharmacists were selected and were assigned to one area each. The pharmacists had at least 4 years' experience of performing medication reviews. Patients eligible for inclusion were users of the multi-dose drug dispensing system aged 75 years or older, living in nursing homes or their own homes with municipally provided home care. Patients were included in the study after they provided written consent (directly or through relatives in cases of severe cognitive impairment). The patients were included between 1 September and 16 December 2011 with follow-up data collection continued until 16 February 2012. An overview of the actions in the study is presented in Fig. 1.

3.2 Implementation

Prior to the patient's annual visit and medication renewal by the GP, nurses collected the patient's written consent for participation in the study and conducted a specific symptom evaluation and health status check including blood pressure, pulse, weight, tendency to fall and confusion, using a validated symptom assessment form (Phase-20) [20]. After inclusion, the pharmacist used closed, non-transparent envelopes to randomise the patient to one of two groups: control or intervention (Fig. 2). The randomisation was performed using a random number generator and stratified only for geographic area. Medication lists (MDD cards) were printed by the pharmacists who had received permission to access patients' EMR as well as the electronic MDD record.

Fig. 1 Overview of the study.
DRPs drug-related problems



3.3 Intervention

For patients in the intervention group the pharmacists performed a systematic medication review without personal patient contact. The medication review included assessment of relevant parts of the EMR and collection of data on the patient's blood sample results for creatinine, estimated glomerular filtration rate (eGFR), cystatin C, haemoglobin, sodium and potassium plasma levels.

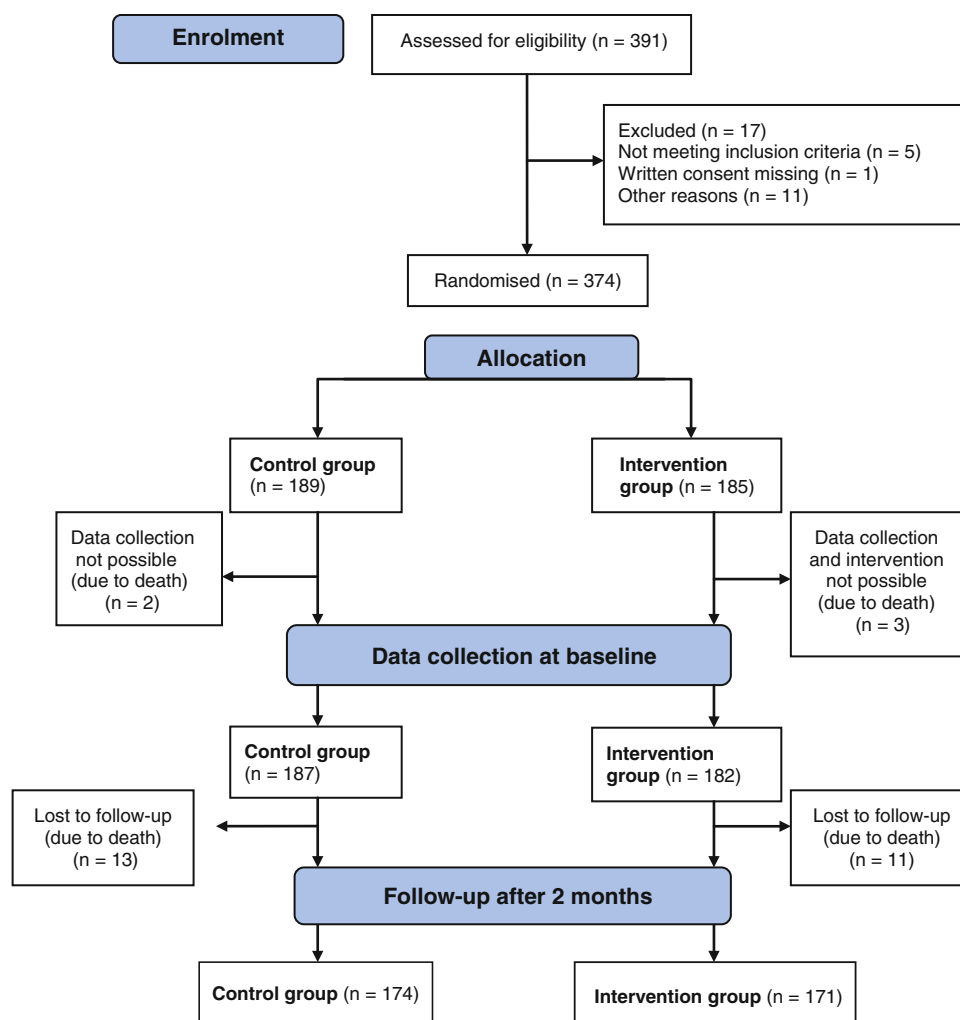
To identify DRPs the clinical pharmacist initiated medication reviews based on the background information (symptom assessment form and the MDD cards). The

working process was carried out in a structured way with formularies compiled from the LImm model [18].

The following predetermined risk categories for identifying DRPs were taken into account by the pharmacist and documented by the student:

- Drugs that required therapeutic monitoring
- Inappropriate drugs for elderly according to The National Board of Health and Welfare (PIMs)
- Drugs that are not recommended according to the regional drug and therapeutics committee
- Problems with administration/handling of the drugs (crush, cut, inhalation technique)
- C/D drug-drug interactions (C interactions are those involving a drug combination that could require dose

Fig. 2 CONSORT study flow chart of the inclusion and assessment process



adjustment; D interactions are those involving a drug combination that ought to be avoided)

- Drug type or drug dosage not adjusted for the patient (renal function, liver function)
- Unclear indication for drug treatment
- Suboptimal treatment
- Drugs causing potential adverse drug reaction.

The check list including the nine risk categories was an instrument to facilitate the medication review.

PIMs were identified according to the national guidelines of the Swedish National Board of Health and Welfare regarding drug therapy in the elderly [7].

The pharmacists' recommendations were documented in patients' EMRs. The feedback to the physician varied depending on the PHCC's routines and organisation and consisted of team rounds, written contact, personal contact and telephone contact.

To ensure that the pharmacists worked similarly, they were formally instructed in one tutorial by the head pharmacist (E.R.) about the method of medication review, had

monthly meetings with the data collector (S.W.) and had one meeting with the head researcher (V.M.). In addition, the head pharmacist was available for consultation throughout the entire study.

3.4 Data Collection and Statistical Analysis

The required sample size was estimated to be at least 160 patients ($n = 80$ per group) by power calculation analysis ($p = 0.05$; power: 0.80) based on the assumption that 40 % of respondents would have at least one inappropriate drug. The intervention was expected to reduce this proportion to 20 %. The calculation was based on previous studies on drug consumption in the elderly [8].

For the intervention patients, S.W. and V.M. compiled drugs associated with the DRPs and assigned categories of risk and type of suggested change in collaboration with the consulting research pharmacist (Å.B.). Medication lists were not assessed for DRPs for the control patients for ethical reasons.

During the data collection, medication lists and patients' EMRs were reviewed at baseline and after 2 months. Drugs were classified according to the Anatomical Therapeutic Chemical (ATC) classification system [21].

The documented DRPs were further classified by S.W and V.M into the seven categories used by Cipolle, Strand and Morley [22]: need for additional therapy, unnecessary drug therapy, wrong drug, dosage too low, adverse drug reaction, dosage too high and compliance problems.

Both S.W. and V.M. participated in the ongoing review meetings of the research team, where the input method was discussed continuously, in order to assure the quality of the collected data.

If a drug prescription was for both continuous use and as needed, it was counted as one drug. Drugs for topical use such as eye drops, moisturisers and topical steroids were included; short-term antibiotic prescriptions were not.

The primary outcome measure was change in the proportion of patients taking PIMs [7] including one or more of the following drugs: intermediate- or long-acting benzodiazepines (ATC group N05BA01, N05CD02 and N05CD03), antipsychotics [N05A, excluding lithium (N05AN)], tramadol (N02AX), propiomazine (N05CM) and drugs with anticholinergic effects (R06, G04 and N05BB). Secondary outcome measures were percentage of patients taking ten or more medications (regularly or as needed) and percentage of patients taking three or more psychotropic drugs (from one or more of the following ATC groups: N05A, N05B, N05C and N06A) regularly or as needed before and after the intervention. The secondary outcome measures are based on the definition of "polypharmacy" as described by the Swedish National Board of Health and Welfare.

Intermediate- and long-acting benzodiazepines prescribed in Sweden are nitrazepam, flunitrazepam and diazepam.

The average age and sex distribution of the patients were determined, as were the average number of drugs per patient and the proportion of patients using drugs in the different ATC subgroups. Data on DRPs, if recommended changes were performed or not and actions taken were also collected. Identification of DRPs was a part of the intervention and thus not made in the control group. The DRPs were identified based on the symptom assessment performed by the nurse at baseline. This was not repeated after the intervention. Focus was on the medication changes in the medication lists with data collection before and after the medication reviews.

Data were analysed according to the "intention-to-treat" principle with the last value carried forward using a single imputation method [23]. A significance level of $\alpha = 0.05$ was used. Statistical tests were performed for both intention-to-treat and per-protocol analyses using Student's

t test and McNemar's test for pairwise observations using IBM SPSS version 20.0 UK.

4 Results

A flow chart of the inclusion and assessment steps is presented in Fig. 2. Baseline characteristics are presented in Table 1. In the intervention group the pharmacist had a face-to-face encounter with the physician during team sessions in 20 % of cases. Distance medication reviews were performed in 80 % of the cases. The control and intervention groups were similar, and a majority of patients were females and lived in nursing homes.

4.1 PIMs

A total of 391 patients were assessed, and 369 were included in the intention-to-treat analysis. The proportion of patients with at least one PIM decreased in the intervention group (by 6 %; $p = 0.007$) but not in the control group ($p = 1.0$) (Table 2). Similarly, the number of patients taking ten or more drugs decreased in the intervention group but not in the control group (Table 2). No differences in mortality between the groups were seen after the medication reviews: 6.8 % of patients in the control group and 5.9 % of patients in the intervention group died during follow-up (Fig. 2). Nearly one-third of the patients in both the control and intervention groups had at least one PIM for elderly patients at baseline (Table 2). The total number of drugs and number of continuous drugs decreased significantly between baseline and follow-up in the intervention group (Table 3). No significant decreases after the medication reviews were noted in the medication subgroups (antipsychotics, benzodiazepines, etc.). Similar results were found in both intention-to-treat and per-protocol analyses.

4.2 DRPs

DRPs were identified in 93 % of the 182 patients in the intervention group. The total number of DRPs in this group was 431 with a mean of 2.5 DRPs per patient [range 0–9 (SD = 1.5)]. No difference between the number of DRPs in community-dwelling patients [mean 2.55 (SD = 1.29)] and nursing home patients [mean 2.53 (SD = 1.33)] was seen ($p = 0.767$).

Drugs acting on the nervous system (26 %), cardiovascular system (25 %) and blood and blood-forming organs (15 %) were the most common ATC classes involved in DRPs.

The distribution of the seven main categories of DRPs identified when data were collected is shown in Fig. 3. Of

Table 1 Baseline characteristics of intervention and control group patients

Characteristic	Control group	Intervention group	<i>p</i> value
Female, <i>n</i> (%)	142 (75.9)	138 (75.8)	0.980 ^a
Age, mean (SD)	87.7 (5.5)	87.0 (5.8)	0.662 ^b
Place of residence, <i>n</i> (%)			
Community	47 (25.1)	43 (23.6)	0.736 ^a
Nursing home	140 (74.9)	139 (76.4)	
No. of drugs, mean (SD)	12.1 (4.7)	11.4 (4.2)	0.903 ^b
No. of continuous drugs, mean (SD)	9.7 (3.9)	9.3 (3.7)	0.528 ^b
No. of drugs as needed, mean (SD)	2.2 (1.8)	2.1 (1.7)	0.399 ^b
No. of antipsychotics, mean (SD)	0.11 (0.36)	0.14 (0.35)	0.137 ^b
No. of intermediate- or long-acting benzodiazepines, mean (SD)	0.06 (0.25)	0.10 (0.29)	0.070 ^b
No. of anticholinergics, mean (SD)	0.12 (0.34)	0.08 (0.26)	0.040 ^b
No. of propiomazine, mean (SD)	0.04 (0.19)	0.04 (0.19)	0.918 ^b
No. of tramadol, mean (SD)	0.06 (0.24)	0.07 (0.27)	0.873 ^b
No. of psychotropics, mean (SD)	1.93 (1.37)	1.71 (1.37)	0.750 ^b

SD standard deviation

^a Chi-square test^b Student's *t* test

the identified DRPs, 67 (16 %) were related to PIMs, as follows: antipsychotics (27), intermediate- or long-acting benzodiazepines (15), tramadol (11), anticholinergics (9) and propiomazine (5).

The two most common intervention recommendations the pharmacist presented to the physician were withdrawal of drug therapy (30 %) and reduced dosage (28 %) (Fig. 4).

Fifty-six percent (241) of the presented DRPs (Fig. 4) resulted in actions taken by the physician such as changes in medication, with a minimum of one and maximum of seven changes for the same patient [mean 1.44 (SD = 1.33)] with no difference between the community-dwelling and the nursing home patients ($p = 0.946$) or between the group receiving team-based medication reviews compared to the distance medication reviews ($p = 0.363$).

The changes in the actions taken by the physician regarding PIMs were significant ($p = 0.003$) for “lowered dosage” (Table 4) and there was a clear tendency to withdraw the PIMs, although it was not significant. There were no significant differences in actions taken on PIMs

between the group receiving team-based medication reviews and the group receiving distance medication reviews.

5 Discussion

Our study showed that medication reviews involving pharmacists in primary health care reduced the number of patients with PIMs.

The majority of the patients in the present study were women, were living in nursing homes, were old and were using a large number of drugs, characteristics similar to those in other studies [24–26]. The results demonstrate that the assessed care model led to a reduction in the number of intervention group patients taking PIMs and the total number of drugs these patients were taking and identified common DRPs [27–29], such as overprescribing or unclear reasons for medication use.

Similar to another study using a multidisciplinary approach [30], the present study did not show a decrease in

Table 2 Changes in number of patients with PIMs, patients with ≥ 10 drugs or ≥ 3 psychotropic drugs

Variable	Group	Frequency at randomisation (%)	Frequency at follow-up (%)	<i>p</i> value ^a
No. of patients with ≥ 10 drugs	Control group	123 (65.7)	120 (64.1)	0.549
	Intervention group	120 (65.9)	107 (58.7)	0.001
No. of patients with ≥ 3 psychotropics	Control group	60 (32.0)	64 (34.2)	0.219
	Intervention group	47 (25.8)	49 (26.9)	0.754
No. of patients with PIMs	Control group	58 (31.1)	57 (30.5)	1.000
	Intervention group	60 (33.0)	49 (27.0)	0.007

PIM potentially inappropriate medication

^a McNemar's test

Table 3 Changes in medication in the control and intervention groups at follow-up

	Group	Mean number of drugs (range) at baseline	Mean number of drugs (range) at follow-up	<i>p</i> value ^a
No. of drugs	Control	12.1 (3–28)	12.1 (3–29)	0.782
	Intervention	11.4 (2–21)	10.8 (0–22)	<0.001
No. of continuous drugs	Control	9.7 (1–27)	9.6 (1–25)	0.327
	Intervention	9.3 (1–20)	8.8 (1–18)	<0.001
No. of drugs as needed	Control	2.2 (0–12)	2.5 (0–12)	0.061
	Intervention	2.1 (0–10)	2.0 (0–8)	0.171
No. of antipsychotics ^b	Control	0.11 (0–3)	0.11 (0–3)	1.000
	Intervention	0.14 (0–1)	0.13 (0–1)	0.158
No. of intermediate- or long-acting benzodiazepine ^c	Control	0.06 (0–2)	0.06 (0–2)	1.000
	Intervention	0.10 (0–1)	0.10 (0–1)	0.556
No. of anticholinergics ^d	Control	0.12 (0–2)	0.10 (0–3)	0.319
	Intervention	0.08 (0–1)	0.08 (0–1)	1.000
No. of propiomazine ^e	Control	0.04 (0–1)	0.04 (0–1)	1.000
	Intervention	0.04 (0–1)	0.03 (0–1)	0.416
No. of tramadol ^f	Control	0.06 (0–2)	0.07 (0–1)	0.416
	Intervention	0.07 (0–2)	0.04 (0–1)	0.103
No. of psychotropics ^g	Control	1.93 (0–6)	1.96 (0–6)	0.224
	Intervention	1.71 (0–6)	1.69 (0–6)	0.082

Anatomical Therapeutic Chemical (ATC) classification system codes for medications are provided in footnotes b–g

^a Student's *t* test

^b N05A excluding lithium (ATC code N05AN)

^c N05BA01, N05CD02 and N05CD03

^d R06, G04 and N05BB

^e N05CM

^f N02AX

^g N05A, N05B, N05C and N06A

the number of patients taking three or more psychotropics, possibly because of multiple illnesses and the remaining need for psychotropics due to cognitive or other psychiatric impairments in this group of patients. This conclusion is, however, only speculative and future research is required to explore a possible association between elderly patients' multiple use of drugs affecting the nervous system and psychiatric morbidity.

There was no difference in mortality after the performed medication reviews between the intervention and control groups, but the short follow-up period and multiple illnesses in this frail group of elderly patients should be taken into consideration.

Assessing the effect of optimisation strategies on the appropriateness of prescribing in elderly patients is currently a priority for both clinical and interventional health care research [31]. Multi-disciplinary approaches have been highlighted [32], although according to a meta-analysis of randomised controlled trials of pharmacist-led medication reviews, the evidence of an impact on clinical outcomes (e.g. morbidity) and health-care use was inconclusive [33].

The analysis excluded, however, interventions delivered by combinations of health professionals (e.g. physician and nurses) where the pharmacist was only partly involved. This accentuates the difficulties in measuring the effects of such interventions. Although there is broad knowledge of medication use in older people and tools exist to improve adherence to treatment guidelines, the prevalence of inappropriate prescribing remains high and further studies are needed to identify effective interventions [34].

A strength of our study is that the pharmacists were blinded to patient allocation but not blinded performing the medication reviews. The DRPs were identified by symptom assessment by a nurse working closely with the patient. This information was included by the pharmacist in the written feedback to the physician that was recorded in the patient's EMR and also faxed to the physician as a reminder regardless of medication review form. The MDD cards and EMRs were the central instruments for the assessment of drug therapy, giving current information to the pharmacist and responsible physician and therefore increasing the ability of pharmacists to make an accurate

Fig. 3 Distribution of drug-related problems in the intervention group by category (N = 431)

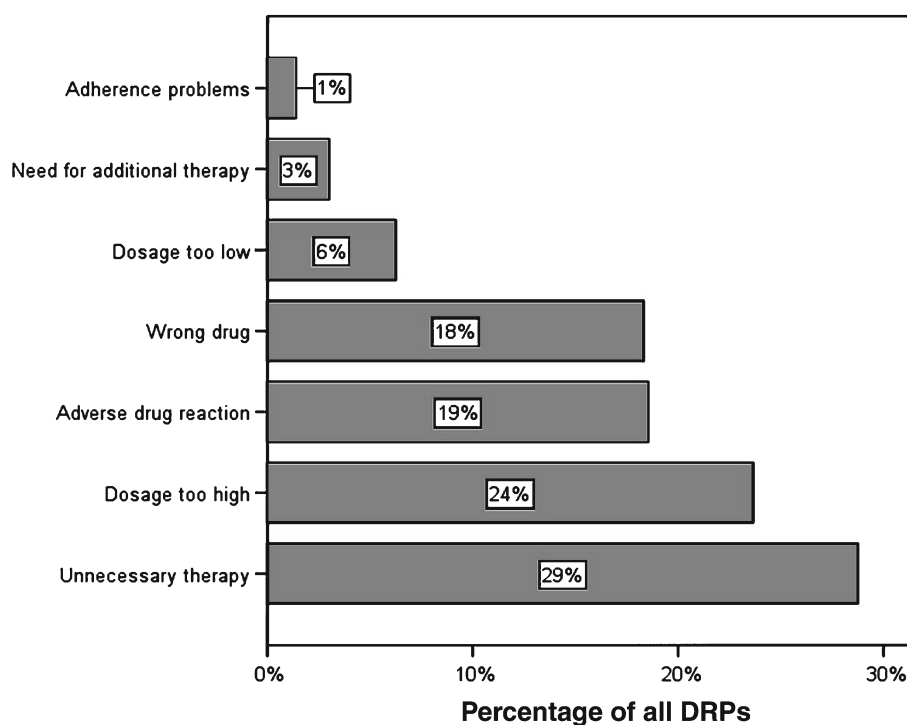
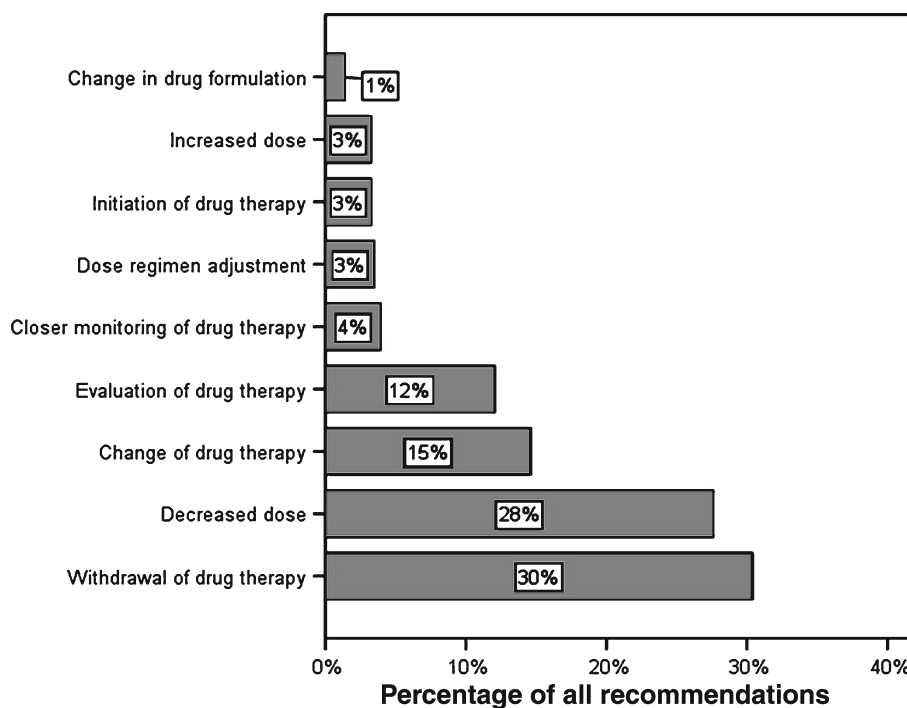


Fig. 4 Distribution of recommended changes in drug therapy in the intervention group (N = 431)



decision in recommending changes in medication. No other medication prescribing interventions were conducted in the districts at the time of the study that impacted on the results. Physicians' decision-making in medication changes was not influenced by patients' living form, implying that the present model of medication review could be applicable

in both community-dwelling and nursing home patients with similar results.

The results from this study have to be interpreted with acknowledgement of its limitations. The pharmacists did not have any direct contact with the assessed patients. Therefore, the identified DRPs are only potential DRPs.

Table 4 Frequency of changes in PIMs in the control group versus the intervention group

Action taken by the physician on drug therapy	No. of cases (percent)		<i>p</i> value ^a
	Control	Intervention	
Dose not changed	56 (76.8)	45 (64.8)	0.349
PIM out	8 (11.5)	13 (17.5)	0.238
New PIM in	7 (10.1)	2 (2.7)	0.098
Lowered dose	0 (0.0)	10 (13.5)	0.003
Increased dose	1 (1.4)	1 (1.3)	0.995

PIM potentially inappropriate medication

^a Student's *t* test

Feedback between the pharmacists and the physicians varied from team discussions to distance reviews, which may partly explain the low rate of physician response in performing medication changes. Fifty-six percent of the presented suggestions led to medication changes. These figures are low compared to those for team-based interventions including a responsible physician in secondary care (65–90 %) [35, 36]. In a British study of elderly nursing home patients, 75 % of the pharmacist's proposals were accepted and of these 76 % were implemented [37]. The present study assessed the implemented medication changes, with results similar to those from other studies performed in primary care [38, 39]. The medical literature supports the theory that valid clinical care recommendations do not always have the desired impact on physicians' behaviour due to cultural barriers [40, 41] or contextual factors (e.g. staffing and resources) [42]. Our study shows that the physicians responded in similar ways after the distance medication reviews compared to the team-based medication reviews.

Inter-professional medication reviews with pharmacists are often studied when performed in face-to-face team discussions [16, 27]. Despite this, distance reviews can have benefits such as accessibility despite large geographical patient distributions and have been performed in southern Sweden as an alternative to team-based medication reviews with positive results concerning quality and quantity of medication and drug costs [43]. Comparison with a model using team-based reviews in another Swedish region showed similar results and the possibility to implement the method should be taken into account in order to improve physicians' adherence to drug therapy guidelines and the inter-professional collaboration.

Medication reviews as interventions performed by pharmacists not primarily responsible for the prescribing decision have previously been criticised for not delivering clear positive outcomes or even potentially worsening health outcomes [44]. Despite this, the present study shows

an effect on the primary outcome measure (number of patients with PIMs).

We estimated that physicians might be most prone to take action within 2 months after the medication review. A longer period to follow-up might also risk a larger dropout because of death in this group of frail patients.

However, the 2-month follow-up period after the intervention may have been too short to measure withdrawal of psychotropic drugs that need a slow reduction in dosage. The analysis of the actions taken by physicians showed a significantly higher frequency of PIM dosage reduction in the intervention group compared to the control group. Dosage reduction is a preferable and recommended step when withdrawal of psychotropics such as long-acting benzodiazepines or antipsychotics is planned.

It is important to mention that the pharmacist's role in reviewing the medication list must be weighed against the clinical reasoning in the final patient assessment and that the path from medication review to the actual implementation of the proposed changes is a complex process. This process starts with the nurses' observation and ends up with the physician's decision.

The assessed method addressed the complexity of prescribing in the elderly, where the professionals were able to collaborate and where use of information technology tools improved drug therapy.

Health outcomes such as improvement in quality of life or effect on hospital admissions were not investigated in this study but should be considered in future studies in order to demonstrate the effectiveness of this kind of intervention.

6 Conclusions

This study verifies that inappropriate prescribing is a problem in Swedish elderly patients living in the community or nursing homes, mirroring the results of international studies [26]. Medication reviews involving pharmacists in primary health care appear to be a feasible method to reduce the number of patients with PIMs, thus improving the quality of pharmacotherapy in elderly patients.

Acknowledgments The authors especially want to thank the four pharmacists who performed the medication reviews (Annika Dobszai, Karin Fält, Martina Haggren and Krister Karlsson), the pharmacist student who initially collected the data (Susan Wong) and the municipal care nurses. We are indebted to Stephen Gilliver for his expertise and invaluable advice in proofreading the manuscript.

Conflict of interest None declared

Disclaimer The opinions or assertions in this article are the views of the authors and are not to be construed as official or as necessarily

reflecting the views of the Swedish Medical Products Agency, where one of the authors is employed.

Funding The study was conducted with government funding for projects involving improvement of drug therapy in the elderly.

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